

Homework 3
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Q.A) The volume charge density inside an atomic nucleus of radius a is: $\rho_v = \rho_0 \left(\frac{a^3 - R^3}{a^3} \right)$

(a) calculate total charge

(b) Determine E and V outside nucleus.

(c) Determine E and V inside nucleus.

$$(a) \quad Q = \int \rho_v \cdot dv$$

$$\Rightarrow \int_{r=0}^a \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \rho_0 \left(\frac{a^3 - R^3}{a^3} \right) \cdot R^2 \sin\theta \, dR \, d\theta \, d\phi$$

$$\Rightarrow \frac{\rho_0}{a^3} \iiint (a^3 - r^3) r^2 \sin\theta \, d\theta \, d\phi \, dr$$

$$\frac{\rho_0}{a^3} \int_{r=0}^a (a^3 - r^3) r^2 \cdot dr \cdot \phi \Big|_0^{2\pi} \cdot \left(-\cos\theta \right) \Big|_0^{\pi}$$

$$\frac{\rho_0}{a^3} \int_{r=0}^a (a^3 - r^3) r^2 \cdot dr \cdot (2\pi) \cdot (2)$$

$$\frac{4\pi\rho_0}{a^3} \cdot \left[\frac{a^3 r^3}{3} - \frac{r^6}{6} \right]_0^a$$

$$\frac{4\pi\rho_0}{a^3} \left[\frac{a^6}{3} - \frac{a^6}{6} \right] \Rightarrow \frac{4\pi\rho_0}{a^3} \left[\frac{a^6}{6} \right]$$

$$Q = \frac{4\pi\rho_0 a^3}{6} \Rightarrow \frac{2\pi\rho_0 a^3}{3} C$$

b)

$$Q = \oint_S D \cdot dS = \oint_S \epsilon_0 E_r \cdot dS$$

$$\frac{2\pi\rho_0 a^3}{3} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \epsilon_0 E_r a_r \cdot r^2 \sin\theta d\theta d\phi$$

$$= \epsilon_0 E_r r^2 \int_{\phi=0}^{2\pi} d\phi \cdot \int_{\theta=0}^{\pi} \sin\theta d\theta$$

$$\frac{2\pi\rho_0 a^3}{3} = 4\pi\epsilon_0 E_r r^2$$

$$E_r = \frac{\rho_0 a^3}{6\epsilon_0 r^2}, \quad \vec{E} = \frac{\rho_0 a^3}{6\epsilon_0 r^2} \cdot \vec{a}_r$$

$$V = -\int_L E \cdot dl \quad \therefore dl = dr a_r$$

$$V = -\int_{\infty}^r \frac{\rho_0 a^3}{6\epsilon_0 r^2} \cdot dr$$

$$= -\frac{\rho_0 a^3}{6\epsilon_0} \left[-\frac{1}{r} \right]_{\infty}^r = \frac{\rho_0 a^3}{6\epsilon_0 r} = V$$

c)

$$\rho_{enc} = \frac{4\pi\rho_0}{a^3} \left[\frac{a^3 r^3}{3} - \frac{r^6}{6} \right]_0^r$$

$$= \frac{4\pi\rho_0}{a^3} \left[\frac{2a^3 r^3 - r^6}{3} \right]$$

$$\epsilon_0 E_r 4\pi r^2 = \frac{2\pi\rho_0}{3a^3} (2a^3 r^3 - r^6)$$

$$\epsilon_0 E_r 2r^2 = \frac{\rho_0 r^2 (2a^3 - r^3)}{3a^3}$$

$$E_r = \frac{\rho_0 r (2a^3 - r^3)}{6a^3 \epsilon_0}$$

$$\vec{E} = \frac{\rho_0 r (2a^3 - r^3)}{6a^3 \epsilon_0} \cdot \vec{a}_r$$

$$V = -\int_{\infty}^r E \cdot d\ell = -\int_{\infty}^a E_{ext} \cdot d\ell - \int_a^r E_{int} \cdot d\ell$$

$$= -\int_{\infty}^a \frac{\rho_0 a^3}{6\epsilon_0 r^2} \cdot dr - \int_a^r \frac{\rho_0 r (2a^3 - r^3)}{6a^3 \epsilon_0} \cdot dr$$

$$= \frac{\rho_0 a^3}{6\epsilon_0} \left[\frac{-1}{r} \right]_{\infty}^a - \frac{\rho_0}{6a^3 \epsilon_0} \left[\frac{a^3 r^2}{5} - \frac{r^5}{5} \right]_a^r$$

$$= \frac{\rho_0 a^2}{6\epsilon_0} - \frac{\rho_0}{6a^3 \epsilon_0} \left[\frac{a^3 r^2}{5} - \frac{r^5}{5} - \frac{a^5}{5} + \frac{a^5}{5} \right]$$

$$= \frac{\rho_0 a^2}{6\epsilon_0} - \frac{\rho_0}{6a^3 \epsilon_0} \left[\frac{a^3 r^2}{5} - \frac{r^5}{5} - \frac{a^5}{5} + \frac{a^5}{5} \right]$$

$$= \frac{\rho_0}{6\epsilon_0} \left[a^2 - \frac{a^3 r^2}{a^3} - \frac{r^5}{5a^3} - \frac{a^3 r^2}{a^3} + \frac{a^5}{5a^3} \right]$$

$$V = \frac{\rho_0}{6\epsilon_0} \left[\frac{a^2 - r^3 + r^5}{5a^3} + \frac{a^2 - a^2}{5} \right]$$

Q.2B) A thick spherical shell carries charge density

$$\rho_r = \frac{k}{R^3}, \quad a \leq R \leq b$$

Find $\vec{E}$ in;

(a) $R < a$; (b) $a < R < b$; (c) $R > b$.

(a) For $R < a$:

$$Q_{enc} = 0, \quad \vec{D} = 0, \quad \boxed{\vec{E} = 0}$$

(b) For $a < R < b$:

$$Q_{enc} = \oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_r \cdot dV =$$

$$= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} D_r \cdot r^2 \sin\theta d\theta d\phi = \int_V \frac{k}{r^3} \cdot r^2 \sin\theta d\theta d\phi dr$$

$$= 4\pi r^2 \cdot D_r = \int_{r=a}^r \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \frac{k}{r} \sin\theta d\theta d\phi dr$$

$$4\pi r^2 \cdot D_r = 4\pi k (\ln(r) - \ln(a))$$

$$D_r = \frac{k}{r^2} (\ln(r) - \ln(a))$$

$$D_r = \frac{k}{r^2} \ln(r/a)$$

$$\boxed{\vec{E} = \frac{k}{\epsilon_0 r^2} \ln\left(\frac{r}{a}\right) \cdot \hat{a}_r}$$

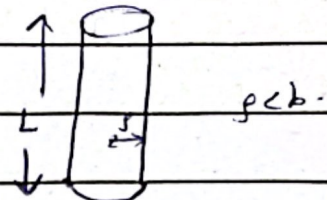
(c) For $R > b$:

$$r = b.$$

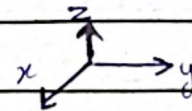
$$\vec{E} = k \frac{\ln\left(\frac{b}{a}\right)}{r^2 \epsilon_0} \vec{a}_r$$

Q.3.c) (a)

$$\rho_v = \begin{cases} \frac{\rho_0 \rho^2}{b^2}, & 0 < \rho < b \\ 0, & b < \rho < a. \end{cases}$$



$$Q_{enc} = \oint_S \epsilon_0 \vec{E} \cdot d\vec{S}$$



LHS:

$$\begin{aligned} &= \oint_S \epsilon_0 \vec{E} \cdot d\vec{S} \\ &= \int_{z=0}^L \int_{\phi=0}^{2\pi} \epsilon_0 E_{ap} \cdot \rho d\phi dz a_{\rho} \end{aligned}$$

$$d\vec{S} = \rho d\phi dz a_{\rho}$$

$$\begin{aligned} &= \epsilon_0 \rho E [z]_0^L [\phi]_0^{2\pi} \\ &= 2\pi L \epsilon_0 \rho E \end{aligned}$$

RHS:

$$\begin{aligned} Q_{enc} &= \int_V \rho_v \cdot dV \\ &= \int_{z=0}^L \int_{\phi=0}^{2\pi} \int_{\rho=0}^b \frac{\rho_0 \rho^2}{b^2} \cdot \rho d\rho d\phi dz. \end{aligned}$$

$$= \frac{\rho_0}{b^2} \cdot \frac{\rho^4}{4} \Big|_0^b \cdot z \Big|_0^L \cdot \phi \Big|_0^{2\pi}$$

$$= \frac{\rho_0 \rho^4 2\pi L}{4b^2}$$

for $0 < \rho < b$.

$$2\pi L \epsilon_0 \rho E = \frac{\rho_0 \rho^4 2\pi L}{4b^2}$$

$$\Rightarrow \vec{E} = \frac{\rho_0 \rho^3}{4\epsilon_0 b^2} a_{\rho}$$

Now $b < \rho < a$ and $0 < \phi < 2\pi$.

LHS: $2\pi \epsilon_0 L \rho E$

RHS: $Q_{enc} = \int_V \rho_v \cdot dV = \int_{z=0}^L \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \rho_v \cdot dV$

$$= \int \int \int \frac{\rho_0 \rho^2}{b} \cdot \rho \, d\rho \, d\phi \, dz$$

$$= \frac{\rho_0 b^2 2\pi L}{4}$$

$$\frac{2\pi \epsilon_0 \rho E}{4} = \frac{\rho_0 b^2 2\pi L}{4}$$

$$\boxed{\bar{E} = \frac{\rho_0 b^2}{4\epsilon_0 \rho} \cdot a \rho} \quad \text{for } \rho > b.$$

(b) It is given that $\bar{E} = 0$ outside cylinder at $\rho \geq a$.

we have $D_{1n} - D_{2n} = \rho_s$ or $\bar{D}_{1n} - \bar{D}_{2n} = \rho_s a n$

$$\bar{E}_2 = \frac{\rho_0 b^2}{4\epsilon_0 \rho} \cdot a \rho \bigg|_{\rho=a} = \frac{\rho_0 b^2}{4a\epsilon_0} \cdot a \rho$$

$$\epsilon_1 E_1 - \epsilon_2 E_2 = \rho_s$$

$$\boxed{\rho_s = -\epsilon_0 E_2 \bigg|_{\rho=a} = \frac{-\rho_0 b^2}{4a}}$$

Q.40) $P = \rho_0 \sqrt{r} \hat{a}_r$. Find $\rho_{ps} = P \cdot \hat{a}_N$ and $\rho_{pv} = -\nabla \cdot P$

$$\rho_{ps} = \rho_0 \sqrt{r} \hat{a}_r \hat{a}_r$$

$$\boxed{\rho_{ps} = \rho_0 \sqrt{r}}$$

$$\rho_{pv} = -\nabla \cdot P$$

$$= -\frac{1}{r} \frac{\partial}{\partial r} (\rho_0 \sqrt{r} \cdot r)$$

$$= -\rho_0 \frac{\partial}{\partial r} (r^{3/2})$$

$$= -\rho_0 \cdot \frac{3}{2} r^{1/2}$$

$$\boxed{\rho_{pv} = \frac{-3\rho_0 \sqrt{r}}{2r} = \frac{-3\rho_0}{2\sqrt{r}}}$$

Q.5E) $\vec{J} = 400 \sin \theta / (r^2 + 4) \hat{R}$ A/m². Find total current flowing through that portion of spherical surface $R=1$, bounded by $0.1\pi < \theta < 0.3\pi$, $0 < \phi < 2\pi$.

$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}, \quad d\vec{S} = r^2 \sin \theta d\theta d\phi \hat{a}_r$$

$$= \int_S \frac{400 \sin \theta}{r^2 + 4} \cdot r^2 \sin \theta d\theta d\phi$$

$$= \frac{400}{5} \int_{\phi=0}^{2\pi} \int_{\theta=0.1\pi}^{0.3\pi} \sin^2 \theta \cdot d\theta d\phi$$

$$I = 80 \cdot 2\pi \cdot \int_{\theta=0.1\pi}^{0.3\pi} \frac{1 - \cos 2\theta}{2} \cdot d\theta$$

$$I = 160\pi \left[\left(\frac{1(0.3\pi)}{2} - \frac{1(0.1\pi)}{2} \right) - \left(\frac{\sin 2(0.3\pi)}{4} - \frac{\sin 2(0.1\pi)}{4} \right) \right]$$

$$I_{enc} = 112.26 \text{ A}$$

Q.6 F) The plates of a capacitor are separated by a distance d and grounded. The dielectric b/w the plates has permittivity ϵ and a charge density $\rho_v(x) = \rho_0 x(x-d)$. Find V and E between the plates of capacitor.

$$\nabla^2 \cdot V = \frac{-\rho_v}{\epsilon}$$

$$\frac{\partial^2 V}{\partial x^2} = \frac{-\rho_0 x(x-d)}{\epsilon}$$

$$\frac{\partial^2 V}{\partial x^2} = \frac{-\rho_0 x^2}{\epsilon} + \frac{\rho_0 x d}{\epsilon}$$

$$\frac{\partial V}{\partial x} = \frac{-\rho_0 x^3}{3\epsilon} + \frac{\rho_0 x^2 d}{2\epsilon} + C_1$$

$$V(x) = \frac{-\rho_0 x^4}{12\epsilon} + \frac{\rho_0 x^3 d}{6\epsilon} + C_1 x + C_2$$

$$\underline{x=0, V(x)=0}$$

$$0 = C_2$$

$$\underline{x=d, V(x)=0}$$

$$0 = \frac{-\rho_0 d^4}{12\epsilon} + \frac{\rho_0 d^4}{6\epsilon} + d C_1$$

$$c_1 = \frac{\rho_0 d^3}{12\epsilon} + \frac{\rho_0 d^3}{6\epsilon} = \frac{3\rho_0 d^3}{12\epsilon} = \frac{\rho_0 d^3}{4\epsilon}$$

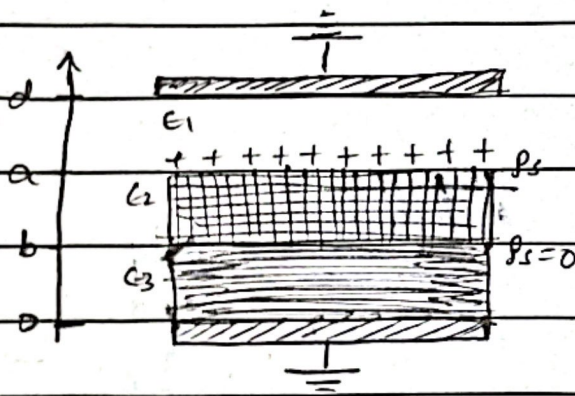
$$V(x) = -\frac{\rho_0 x^4}{12\epsilon} + \frac{\rho_0 x^3 d}{6\epsilon} + \frac{\rho_0 x d^3}{4\epsilon}$$

$$E = -\nabla \cdot V$$

$$= -\frac{\partial}{\partial x} \left(-\frac{\rho_0 x^4}{12\epsilon} \right) + \frac{\partial}{\partial x} \left(\frac{\rho_0 x^3 d}{6\epsilon} + \frac{\rho_0 x d^3}{4\epsilon} \right)$$

$$\vec{E} = -\frac{\rho_0 x^3}{3\epsilon} + \frac{\rho_0 x^2 d}{2\epsilon} + \frac{\rho_0 d^3}{4\epsilon} \cdot a_x$$

Q.7 G)



Find $\vec{E}$ in all three regions.

$$V_1(x) = A_1 x + B_1 ; a \leq x \leq d$$

$$V_2(x) = A_2 x + B_2 ; b \leq x \leq a$$

$$V_3(x) = A_3 x + B_3 ; 0 \leq x \leq b$$

Electric fields are obtained are;

$$\vec{E}_1 = -\frac{\partial V_1}{\partial x} \cdot a_x = -A_1 a_x$$

$$\vec{E}_2 = -\frac{\partial V_2}{\partial x} a_x = -A_2 a_x$$

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Boundary conditions yield:

$$x=d, V_1=0$$

$$x=a, V_1=V_2, \epsilon_1 E_1 - \epsilon_2 E_2 = \rho_s$$

$$x=b, V_2=V_3, \epsilon_2 E_2 - \epsilon_3 E_3 = 0$$

$$x=0, V_3=0$$

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$$\text{Eq (1)} \Rightarrow A_1 d + B_1 = 0$$

$$\text{Eq (2)} \Rightarrow A_1 a + B_1 = A_2 a + B_2$$

$$\text{Eq (3)} \Rightarrow A_2 b + B_2 = A_3 b + B_3$$

$$\text{Eq (4)} \Rightarrow A_3 (0) + B_3 = 0$$

$$\text{Eq (5)} \Rightarrow -\epsilon_1 A_1 + \epsilon_2 A_2 = \rho_s$$

$$\text{Eq (6)} \Rightarrow -\epsilon_2 A_2 + \epsilon_3 A_3 = 0$$

Solving the above we obtain:

$$A_1 = -\frac{\rho_s}{P_1} \frac{P_2}{P_1}$$

$$A_2 = -\frac{a \epsilon_3 \rho_s - d \epsilon_3 \rho_s}{P_1}$$

$$A_3 = -\frac{\epsilon_2 \rho_s (a-d)}{P_1}$$

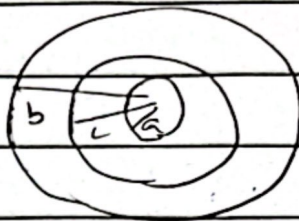
$$B_1 = \frac{d \rho_s}{P_1} \frac{P_2}{P_1}$$

$$B_2 = -\frac{b \rho_s (a \epsilon_2 - a \epsilon_3 - d \epsilon_2 + d \epsilon_3)}{P_1}$$

where $P_1 = a e_1 e_3 - a e_2 e_3 + b e_1 e_2 - b e_1 e_3 + d e_2 e_3$.

$$P_2 = a e_3 + b e_2 - b e_3$$

Q.8H) The cross section of a spherical capacitor is shown. Find capacitance when $a = 2\text{mm}$, $b = 6\text{mm}$ and $c = 4\text{mm}$, $\epsilon_{r1} = 3$ and $\epsilon_{r2} = 4$.



$$C = 4\pi\epsilon \frac{1}{\frac{1}{a} - \frac{1}{b}}, \quad b > a$$

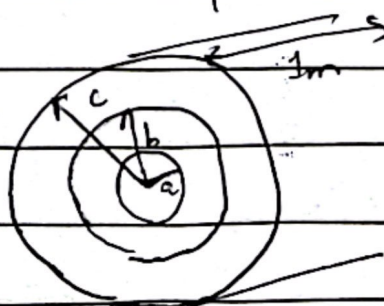
$$C_1 = 4\pi\epsilon_0\epsilon_{r1} \frac{1}{\frac{1}{c} - \frac{1}{b}}, \quad C_2 = 4\pi\epsilon_0\epsilon_{r2} \frac{1}{\frac{1}{a} - \frac{1}{c}}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$C_{eq} = \frac{\frac{4\pi\epsilon_0\epsilon_{r1}}{\frac{1}{c} - \frac{1}{b}} \times \frac{4\pi\epsilon_0\epsilon_{r2}}{\frac{1}{a} - \frac{1}{c}}}{\frac{4\pi\epsilon_0\epsilon_{r1}}{\frac{1}{c} - \frac{1}{b}} + \frac{4\pi\epsilon_0\epsilon_{r2}}{\frac{1}{a} - \frac{1}{c}}}$$

$$= \frac{16\pi^2\epsilon_0^2\epsilon_{r1}\epsilon_{r2}}{4\pi\epsilon_0 \left(\frac{\epsilon_{r1}}{\frac{1}{a} - \frac{1}{c}} + \frac{\epsilon_{r2}}{\frac{1}{b} - \frac{1}{c}} \right)}$$

Q.9I) calculate resistance of conductor



$$a = 10\text{mm}$$

$$b = 12\text{mm}$$

$$c = 15\text{mm}$$

the conductivities of aluminium, copper and steel are 3.5×10^7 , 5.7×10^7 and 1×10^7 .

$$\nabla^2 V = \frac{\partial^2 V}{\partial z^2} = 0$$

$$\frac{\partial V}{\partial z} = C_1$$

$$V = C_1 z + C_2$$

$$V(z=1) = V_0$$

$$\hookrightarrow V_0 = C_1 + C_2$$

$$C_1 = V_0 - C_2$$

$$V(z=0) = 0$$

$$0 = C_2$$

$$C_1 = V_0, \quad V_0 = V_0$$

$$E = -\nabla V$$

$$= -\frac{\partial V}{\partial z} \cdot a_z = -V_0 a_z$$

$$I = \int_S \sigma \vec{E} \cdot d\vec{S} \quad \therefore d\vec{S} = -\rho d\rho d\phi a_z$$

$$= \sigma \int (-V_0 a_z) (-\rho d\rho d\phi a_z)$$

leaving current

$$= \sigma V_0 \int_{\phi=0}^{2\pi} \int_{\rho=0}^{10\text{mm}} \rho d\rho d\phi \quad (\text{for steel})$$

$$= (1 \times 10^7) V_0 \int_{\phi=0}^{2\pi} \left. \frac{\rho^2}{2} \right|_{\rho=0}^{10\text{mm}} d\phi = (1 \times 10^7) V_0 (2\pi) \frac{(10 \times 10^{-3})^2}{2}$$

$$= 1000\pi V_0$$

for Aluminum: 2π 12mm

$$I = \oint \oint \rho \, d\rho \, d\phi = (3.5 \times 10^7) V_0 (2\pi) \frac{(12 \times 10^{-3})^2}{2}$$

$$= 5040 V_0 \pi$$

for Copper: 2π 15mm

$$I = \oint \oint \rho \, d\rho \, d\phi = (5.7 \times 10^7) V_0 (2\pi) \frac{(15 \times 10^{-3})^2}{2}$$

$$= 12825 V_0 \pi$$

$$R_{\text{steel}} = \frac{V_0}{I} = \frac{V_0}{1000\pi V_0} = \frac{1}{1000\pi}$$

$$R_{\text{aluminum}} = \frac{V_0}{I} = \frac{V_0}{5040\pi V_0} = \frac{1}{5040\pi}$$

$$R_{\text{copper}} = \frac{V_0}{I} = \frac{V_0}{12825\pi V_0} = \frac{1}{12825\pi}$$

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_{\text{steel}}} + \frac{1}{R_{\text{aluminum}}} + \frac{1}{R_{\text{copper}}}$$

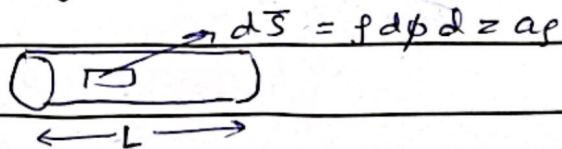
$$\Rightarrow \frac{1}{R_{\text{total}}} = 1000\pi + 5040\pi + 12825\pi$$

$$\Rightarrow \frac{1}{R_{\text{total}}} = 18865\pi \Rightarrow R_{\text{total}} = \frac{1}{18865\pi} \approx 16.9 \mu\Omega$$

Q.10 J)

(a) For $0 < \rho < a$, the space is empty but contains a line charge density ρ_L .

We may comfortably assume that $\vec{E} = E_\rho \hat{a}_\rho$ or $\vec{D} = D_\rho \hat{a}_\rho$



$$\oint \vec{D} \cdot d\vec{S} = Q_{enc} = \rho_L L$$

$$D_\rho \oint d\phi dz = 2\pi L \rho D_\rho = \rho_L L \Rightarrow D_\rho = \frac{\rho_L}{2\pi \rho}$$

$$\vec{D} = D_\rho \hat{a}_\rho$$

$$\vec{E} = \frac{D_\rho}{\epsilon_0} \cdot \hat{a}_\rho = \frac{\rho_L}{2\pi \epsilon_0 \rho} \cdot \hat{a}_\rho \quad \left| \begin{array}{l} \text{since region is free space} \\ \vec{P} = 0 \end{array} \right.$$

(b) $a < \rho < c$:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \therefore \vec{P} = \epsilon_0 \chi_e \vec{E}$$

$$\vec{D} = \epsilon_0 (1 + \chi_e) \vec{E} = \epsilon_0 \epsilon_r \vec{E} = \epsilon_1 \vec{E}$$

$$\vec{D} = \frac{\rho_L}{2\pi \rho} \hat{a}_\rho$$

$$\vec{E} = \frac{\vec{D}}{\epsilon} = \frac{\rho_L}{2\pi \epsilon_1 \rho} \hat{a}_\rho$$

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E} = \frac{\rho_L}{2\pi} \left(1 - \frac{\epsilon_0}{\epsilon_1} \right) \cdot \hat{a}_\rho$$

$$(c) \quad c < r < b$$

$$\bar{D} = \frac{\rho_L a r}{2\pi r} \quad , \quad \bar{E} = \frac{\bar{D}}{\epsilon_2} = \frac{\rho_L a r}{2\pi \epsilon_2 r}$$

$$\bar{P} = \bar{D} - \epsilon_0 \bar{E} = \frac{\rho_L a r}{2\pi r} \left(1 - \frac{\epsilon_0}{\epsilon_2} \right)$$

$$(d) \quad r > b$$

$$\bar{D} = \frac{\rho_L a r}{2\pi r} \quad , \quad \bar{E} = \frac{\rho_L a r}{2\pi \epsilon_0 r}$$

$$\bar{P} = 0 \quad \text{since } r > b \text{ is empty space.}$$